Unit No.4-Tree

Title AVL : Tree Traversal System Using C++

1. Problem Statement

Implement an AVL Tree Traversal System using C++ to construct a self-balancing binary search tree from user-entered integers. The system should perform and display inorder, preorder, postorder, and level-order traversals to understand the structure and arrangement of nodes in the AVL tree.

2. Objectives

- To implement an AVL tree using C++.
- To insert user-entered integers into the AVL tree.
- To maintain the balance of the tree using AVL rotations.
- To perform inorder, preorder, and postorder traversals.
- To implement level-order traversal using a queue.
- To understand different tree traversal techniques.

3. Algorithm

Step 1: Start the program.

Step 2: Create a node structure containing data, left pointer, right pointer, and height.

Step 3: Initialize the root of the AVL tree as NULL.

Step 4: Accept the number of integers from the user.

Step 5: Read the integers one by one and insert them into the AVL tree.

Step 6: Calculate the height and balance factor of each node.

Step 7: If the tree becomes unbalanced, perform the appropriate AVL rotation:

Left-Left (LL) Rotation

Right-Right (RR) Rotation

Left-Right (LR) Rotation

Right-Left (RL) Rotation

Step 8: Perform inorder traversal (Left → Root → Right).

Step 9: Perform preorder traversal (Root → Left → Right).

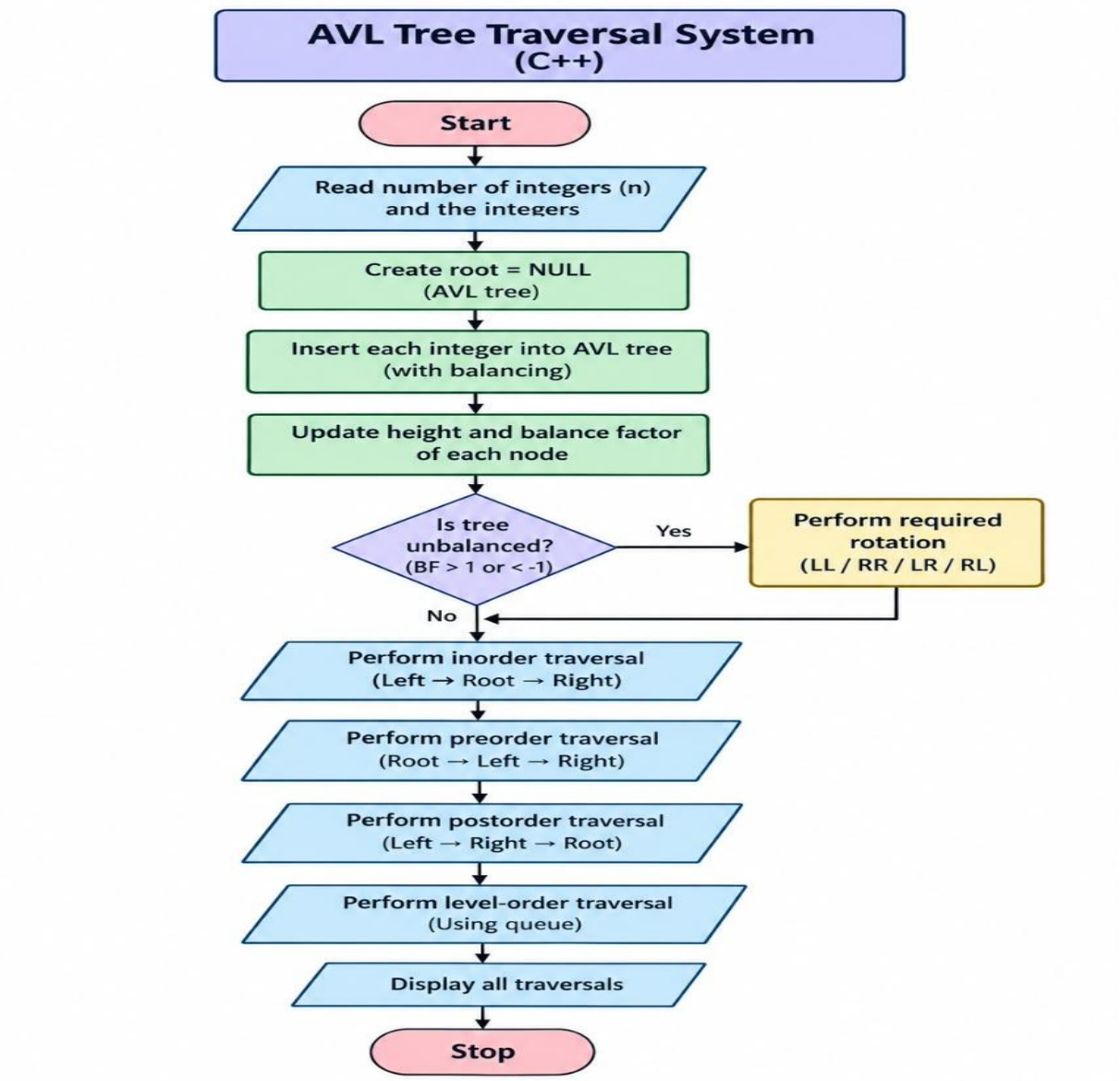
Step 10: Perform postorder traversal (Left → Right → Root).

Step 11: Perform level-order traversal using a queue.

Step 12: Display the results of all four traversals.

Step 13: Stop the program.

4.Flowchart



5.Conclusion

The AVL Tree Traversal System was successfully implemented using C++. The program constructs a self-balancing AVL tree from user-entered integers and performs inorder, preorder, postorder, and level-order traversals. It maintains the balance of the tree using appropriate rotations and uses a queue for level-order traversal. Thus, this application helps in understanding AVL tree operations, balancing techniques, and different tree traversal methods.